Differential Equations Workbook

Higher order equations

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HOMOGENEOUS HIGHER ORDER EQUATIONS

■ 1. Find the general solution to the third order homogeneous differential equation.

$$y''' - 3y'' - y' + 3y = 0$$

■ 2. Find the general solution to the third order homogeneous differential equation.

$$y''' + 6y'' + 8y' = 0$$

■ 3. Find the general solution to the third order homogeneous differential equation.

$$y''' + 15y'' + 75y' + 125y = 0$$

■ 4. Find the general solution to the third order homogeneous differential equation.

$$y''' - 5y'' + 12y' - 8y = 0$$

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■ 5. Find the general solution to the fifth order homogeneous differential equation.

$$y^{(5)} - 2y^{(4)} + 2y''' = 0$$

■ 6. Find the general solution to the seventh order homogeneous differential equation.

$$y^{(7)} + 15y^{(6)} + 125y^{(5)} + 595y^{(4)} + 1{,}795y''' + 2{,}861y'' + 2{,}175y' + 625y = 0$$

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UNDETERMINED COEFFICIENTS FOR HIGHER ORDER EQUATIONS

■ 1. Find the general solution to the third order differential equation.

$$y''' + y'' - 4y' - 4y = 8t^2$$

■ 2. Find the general solution to the third order differential equation.

$$y''' - 6y'' + 9y' - 4y = 1 + e^{2t} + 9e^{4t}$$

■ 3. Find the general solution to the differential equation. Hint:

$x^4 + 4x^3 + 7x^2 + 6x + 2 = (x + 1)^2(x^2 + 2x + 2)$.

$$y'''' + 4y''' + 7y'' + 6y' + 2y = e^{-t}(\cos t + 2\sin t + 4)$$

■ 4. Find the general solution to the differential equation.

$$y''' + 9y'' + 27y' + 27y = 20t^2e^{-3t}$$

■ 5. Find the general solution to the differential equation.

$$y''' + 4y' = \cos(2t) - 1$$

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6. Find the general solution to the differential equation.

$$y'''' + 2y'' + y = t^2 + 8\sin t$$

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VARIATION OF PARAMETERS FOR HIGHER ORDER EQUATIONS

■ 1. Use variation of parameters to find the general solution to the differential equation.

$$y''' - y'' = te^t + 2t + 1 + 3\sin t$$

■ 2. Use variation of parameters to find the general solution to the differential equation.

$$y'' - 5y' + 6y = te^{2t} - e^{3t}$$

■ 3. Use variation of parameters to find the general solution to the differential equation.

$$y''' - y'' - 6y' = t^2e^{3t} - t + 1$$

■ 4. Use variation of parameters to find the general solution to the differential equation.

$$y''' - 6y'' + 5y' + 12y = 10te^{3t} - 4\cos(2t)$$

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■ 5. Use variation of parameters to find the general solution to the differential equation.

$$y''' - y'' + 9y' - 9y = 60\cos(3t)$$

■ 6. Use variation of parameters to find the general solution to the differential equation, given $y(0) = 0$, $y'(0) = -1$, $y''(0) = 10$.

$$y''' + y'' - y' - y = 8e^{-t}$$

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LAPLACE TRANSFORMS FOR HIGHER ORDER EQUATIONS

■ 1. Use the Laplace transform to solve the differential equation, given $y(0) = 0$, $y'(0) = 0$, and $y''(0) = 0$.

$$y''' + 6y'' + 8y' = 8e^{-2t}$$

■ 2. Use the Laplace transform to solve the differential equation, given $y(0) = 0$, $y'(0) = 0$, and $y''(0) = 1$.

$$y''' - 3y' + 2y = 12e^{t} + 24e^{2t}$$

■ 3. Use the Laplace transform to solve the differential equation, given $y(0) = 0$, $y'(0) = 1$, and $y''(0) = 2$.

$$y''' - y'' - 5y' - 3y = 2e^{3t} + 10\sin t$$

■ 4. Use the Laplace transform to solve the differential equation, given $y(0) = 0$, $y'(0) = 0$, and $y''(0) = 0$.

$$y''' - 2y'' + y' - 2y = 5 + 25\sin t$$

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■ 5. Use the Laplace transform to solve the differential equation, given $y(0) = -2$, $y'(0) = 3$, and $y''(0) = 6$.

$$y''' - 6y'' + 12y' - 8y = 27e^{-t} - 2e^{t}$$

■ 6. Use the Laplace transform to solve the differential equation, given $y(0) = 3$, $y'(0) = 0$, and $y''(0) = -9$.

$$y''' + 3y'' + 4y' + 2y = -8\sin t - 6\cos t$$

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SYSTEMS OF HIGHER ORDER EQUATIONS

■ 1. Solve the system of differential equations.

$$x_1' = 2x_1$$

$$x_2' = 4x_3 - x_2$$

$$x_3' = -3x_1$$

■ 2. Solve the system of differential equations.

$$x_1' = 2x_2 - 2x_3$$

$$x_2' = 2x_1 + 4x_2 + 4x_3$$

$$x_3' = -2x_1 + 4x_2 - 3x_3$$

■ 3. Solve the system of differential equations.

$$x_1' = 2x_1 + 4x_3$$

$$x_2' = -x_1 + 2x_2$$

$$x_3' = -x_3$$

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■ 4. Solve the system of differential equations.

$$x_2' = -x_1 + 3x_2 + 5x_3$$

$$x_3' = 2x_1 - 2x_2 + x_3$$

■ 5. Solve the system of differential equations.

$$x_1' = x_1 - x_2$$

$$x_2' = 5x_1 - x_2$$

$$x_3' = -3x_1 + x_2 + 4x_3$$

■ 6. Solve the system of differential equations.

$$x_1' = -x_1 + 3x_2 - x_3$$

$$x_2' = -x_2 + 2x_3$$

$$x_3' = -x_3$$

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SERIES SOLUTIONS OF HIGHER ORDER EQUATIONS

1. Find a power series solution in x to the differential equation.

$$y''' - xy' = 0$$

2. Find a power series solution in x to the differential equation.

$$y''' - y'' = 0$$

3. Find a power series solution in x to the differential equation.

$$xy''' - y' = 0$$

4. Find a power series solution in x to the differential equation.

$$x^2y''' + y'' - xy = 0$$

5. Find a power series solution in x to the differential equation.

$$y''' + xy' + y = 0$$

6. Find a power series solution in x to the differential equation.

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$$y''' + xy'' - y = 0$$

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